



Coláiste na Tríonóide, Baile Átha Cliath
Trinity College Dublin

Ollscoil Átha Cliath | The University of Dublin

Faculty of Science, Technology, Engineering and Mathematics

School of Computer Science & Statistics

Integrated Computer Science
Engineering
Year 4

Semester 1

CSU44031 NEXT GENERATION NETWORKS

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RDS SIM COURT

09.30-11.30

Prof. Nicola Marchetti

Prof. Marco Ruffini

Instructions to Candidates:

Answer ONLY TWO questions from Part A, ONLY TWO questions from part B, ALL questions from part C.

Materials permitted for this examination:

This is a closed book examination. Materials Permitted: Calculator.

PART A [36 marks] - ONLY answer 2 questions**Q.1****[Total: 18 marks]**

Consider the following four schedulers: Round Robin (RR), Maximum Carrier/Interference (Max C/I), Proportional Fair (PF), and Modified Proportional Fair (MPF). The MPF scheduling metric is:

$$n_{max} = \arg \max_n \left(\frac{d_n(t)^\alpha}{r_n(t)^\beta} \right) \quad (1)$$

where $d_n(t)$ is the expected throughput of user n at time t (if the user is scheduled), and $r_n(t)$ is the average past throughput of user n .

- a)** With regard to equation (1) what are the implications for the number of throughput measurements the base station needs to take, to compute the MPF scheduling metric for the cases (i) $\alpha=\beta=0$; (ii) $\alpha=1$ and $\beta=0$; (iii) $\alpha=0$ and $\beta=1$?

[5 marks]

- b)** Would the kind of schedulers do you obtain in the three cases ($\alpha=\beta=0$; $\alpha=1$ and $\beta=0$; $\alpha=0$ and $\beta=1$) favour the achievement of either overall system throughput or fairness, and why?

[5 marks]

- c)** Suppose you are given the following throughput traces for four users U_i , $i=1,\dots,4$, at times $t-3$, $t-2$, $t-1$, and the expected throughput at time t .

	$d_n(t-3)$	$d_n(t-2)$	$d_n(t-1)$	$d_n(t)$
U₁	76.9 Mbps	28.9 Mbps	64.1 Mbps	77.2 Mbps
U₂	18.1 Mbps	90.5 Mbps	10.9 Mbps	11.1 Mbps
U₃	87.7 Mbps	75.7 Mbps	23.2 Mbps	70.9 Mbps
U₄	75.7 Mbps	10.4 Mbps	35.9 Mbps	2.8 Mbps

Which users are going to be scheduled by RR, Max C/I, PF, MPF (with $\alpha=0.1$, $\beta=0.9$) at time t ?

[8 marks]

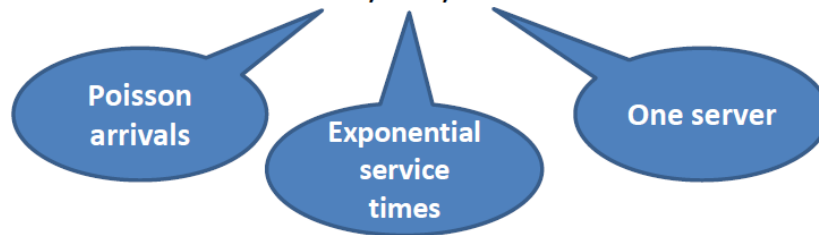
Q.2

[Total: 18 marks]

a) In a single server system, arrivals can be modelled as a Poisson process with rate of 3 min^{-1} and the service times are exponentially distributed with mean 8 s. **Note:** Show your workings when answering the questions below.

- Describe this system using Kendall notation, motivating your choice.

Kendall's notation: $M / M / 1$



- Calculate the average number of customers in the system.
- Calculate the average delay and the average waiting time.
- Calculate the average number of customers in the queue.

[9 marks]

- b)** Calls arrive to a call-centre according to a Poisson process with intensity of 2 calls per minute. The call holding times are exponentially distributed with an average of 12 s. Calls that find all operators busy are blocked.

Note: Show your workings when answering the questions below.

- Describe this system using Kendall notation, motivating your choice.
- How many operators are necessary to keep the blocking probability below 0.01?

[9 marks]

Q.3

In the following equation:

$$C = m \cdot B \log_2 \left(1 + \frac{S}{N} \right) \quad (2)$$

C represents the channel capacity [bps], B the system bandwidth [Hz], S the useful signal power [W], N the noise power [W], and $m = R_s \cdot O_m \cdot R_c$, where R_s is the multiple antenna spatial rate, O_m is the modulation order [bits / modulated symbol], and R_c is the channel coding rate. Assume that $\frac{S}{N} = 1$.

(a) Assume the following system components are available:

(b)

- Multiple antenna schemes: spatial diversity-only mode with a spatial rate of 1, hybrid diversity-spatial multiplexing mode with a spatial rate of 2, spatial multiplexing-only mode with a spatial rate of 3;
- Modulation schemes: BPSK, QPSK, 8-PSK;
- Channel coding rates: 1/4, 1/2, 3/4;
- Bandwidth: 50 MHz, 100 MHz, 150 MHz.

One wants to achieve a throughput equal to 300 Mbps or higher. Outline two systems obtaining that by using the above-mentioned system components.

[5 marks]

(c) Considering equation (2) above, now suppose that an adaptive system can switch among the below configurations A, B and C, where:

- A = {spatial rate = 1, BPSK modulation, channel coding rate = $1/4$ }, if the user is positioned close to the cell edge;
- B = {spatial rate = 2, QPSK modulation, channel coding rate = $1/2$ }, if the user is positioned in an intermediate region between the base station and the cell edge;
- C = {spatial rate = 3, 8-PSK modulation, channel coding rate = $3/4$ }, if the user is positioned close to the base station.

Assuming that a uniform probability distribution for the three user position regions, calculate the average spectral efficiency of the adaptive system.

[7 marks]

(d) In case you want to provide better coverage to mobile users close to the base station, what values (choosing between 'high' and 'low') of coding rate, modulation order and multiple antenna spatial rate should you choose? What will be the impact of the above choice on the system's spectral efficiency? Motivate your answers.

[6 marks]

PART B [36 marks] - ONLY answer 2 questions**Q.4**

- a) With regards to Passive Optical Networks, describe why is the upstream scheduling more complex than downstream scheduling. Then describe how the upstream scheduling works.

[6 marks]

- b) Describe, making appropriate references to the Shannon-Hartley theorem, how the broadband access network has evolved from a pure copper-based telephone network to today's high-speed access.

[6 marks]

- c) Provide a description of three access network sharing techniques.

[6 marks]**Q.5**

Design the following two systems to provide quality of service in a network, based on flow priority.

- a) Provide a solution to the following problem. You need to process four traffic flows:
- Flow 1 has a Committed Information Rate (CIR) of 3 Gb/s and Peak Information Rate (PIR) of 5 Gb/s. Flow 2 has CIR of 9Gb/s and PIR of 12Gb/s, Flow 3 a CIR equals to its PIR of 12Gb/s, and Flow 4 has no CIR and has a PIR of 5 Gb/s.
 - When designing the scheduler, CIR has strict priority over PIR. With regards to flow prioritisation, for CIR the priority is proportional to the CIR threshold. For PIR, packets from Flow 1 and Flow 2 are served at rate two times higher than those for Flow 3 and Flow 4. Calculate the appropriate queue weights, expressed as a percentage, for all CIR and PIR queues.
 - Assuming the physical interfaces have a cost unit of 5 for 25Gb/s and 2 for 10Gb/s, calculate the configuration that satisfies the minimum QoS requirements at the lowest cost (i.e., you need to identify how many interfaces you need and of what data rate).

Show, using appropriate detailed diagrams and explanations, how you would use a chain of different QoS tools to implement your solution. Provide detailed information of any parameters required for any of the QoS tools (you need to indicate the relative weights for all queues, where appropriate).

[9 marks]

b) Provide a solution to the following problem. You need to process two traffic flows:

- Flow 1 has a CIR of 10 Gb/s and PIR of 12 Gb/s. Flow 2 has a CIR of 7Gb/s and a PIR of 9 Gb/s.
- The scheduler should **not** give strict priority to CIR packets, but overall, the sum of all CIR flows should be served at three times the rate of the sum of all PIR flows. Flow 1 CIR packets are served at double rate of Flow 2 CIR packets. And PIR packets are served at the same rate for Flow 1 and Flow 2. Calculate the appropriate queue weights, expressed as a percentage, for all CIR and PIR queues.
- Finally, the system needs to implement an algorithm to drop packets progressively as the load increases. Show the tool that could be used for this purpose, locating it in the right place of the QoS chain.
- Assuming the physical interfaces have a cost unit of 5 for 25Gb/s and 2 for 10Gb/s, calculate the configuration that satisfies the minimum QoS requirements at the lowest cost (i.e., you need to identify how many interfaces you need and of what data rate).

Show, using appropriate detailed diagrams and explanations, how you would use a chain of different QoS tools to implement your solution. Provide detailed information of any parameters required for any of the QoS tools (you need to indicate the relative weights for all queues, where appropriate).

[9 marks]

Q.6

A Wavelength Division Multiplexing (WDM) transmission system uses 20 channels in the C band and needs to operate over a 400 km distance. Amplifiers have 18 dBm of power and noise figure of 8 dB, while the fibre has a nominal attenuation of 0.3 dB/km. The amplifier gain can be set to between 10 and 20 dB. Also consider that a margin of at least 5 dB is required.

- a)** Assuming you need to support an overall capacity of at least 5.8 Tb/s, what is the maximum amplifier distance for the system to work without regenerators? (you can approximate to the closest smallest 5km multiple). Consider that the transceivers have a baud rate of 50 Gbaud and can work at modulation formats of 16QAM, 32 QAM and 64 QAM, which require, respectively, 18, 23 and 28 dB OSNR.

[9 Marks]

- b)** The system uses coherent transmission, so no dispersion compensating fibre (DCF) is required. The transceiver can work at a baud rate of either 33Gbaud (37.5 GHz bandwidth) or 66 Gbaud (75GHz bandwidth). The maximum tolerable dispersion by the digital filter is however 1ns. What is the total capacity you can achieve over the 20 channels if the available OSNR is $> 28\text{dB}$ (assuming the modulation formats as defined in the previous problem)? Assume the average dispersion coefficient of the fibre to be 5 ps/nm/km (assume the following conversion ratio: 100 GHz = 0.8 nm).

[9 Marks]

PART C [28 marks]

For each question you pick, select only ONE answer out of the four available options. Answer ALL questions.

Q.7

Which one among the following functionalities does **not** belong to the OSI Data Link layer?

- a) Medium access
- b) Routing
- c) Error control
- d) Frame synchronisation

[3.5 marks]

Q.8

Which one among the following effects is modelled in the time-selective fading component of a wireless channel model?

- a) Distance between transmitter and receiver
- b) A hill preventing line of sight between transmitter and receiver
- c) Sparks in passing cars
- d) User mobility

[3.5 marks]

Q.9

In the Kendall's notation a/b/m/K, what does 'K' indicate?

- a) Service time distribution
- b) Number of servers
- c) Maximum number of customers allowed
- d) Type of arrival process

Q.10

Which one among the following options, **cannot** be considered a 'meter' for a cognitive radio system?

- a) Transmitted power
- b) Interference level
- c) Signal-to-noise ratio
- d) Bit error rate

[3.5 marks]

Q.11

If a transmission system suffers from nonlinear impairments, which of the following actions would you take to improve performance?

- a) Increase transmission power
- b) Decrease transmission power
- c) Use dispersion shifted fibre
- d) Use multi-mode fibre

[3.5 marks]

Q.12

Which of the following statements is **false** for a chain of amplifiers?

- a) For the same span length, reducing the number of amplifiers increases the OSNR
- b) For the same overall distance a shorter span length gives a higher OSNR
- c) A system with amplifiers with lower noise figures has higher OSNR
- d) A system with a larger number of channels has a higher OSNR

[3.5 marks]

Q.13

Which of the following statements is **false** with respect to broadband access networks?

- a)** Shortening the copper link length provides higher data rate
- b)** Doubling the bandwidth has the same effect as doubling the signal to noise ratio.
- c)** Typically higher capacity copper transmission technology uses higher bandwidth
- d)** Vectoring can double the capacity of a copper transmission system

[3.5 marks]

Q.14

What is the scope of the Weighted Random Early Discard technique?

- a)** To reduce the number of TCP timeouts, while differentiating drop rate among different flows
- b)** To increase fairness among a number of CIR flows
- c)** To maximise the peak rate of TPC flows
- d)** To minimise the peak rate of TPC flows

[3.5 marks]